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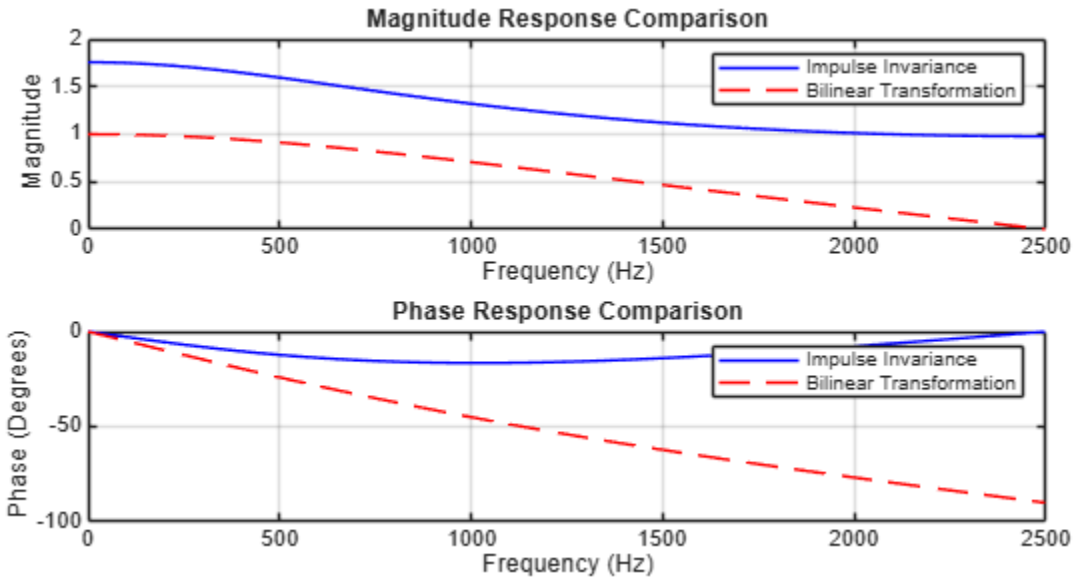
..... 1

```
clear; clc; close all;

Fs = 5000;
T = 1/Fs;
fc = 1000;
Omega_c = 2*pi*fc;

% Analog filter
b_a = Omega_c;
a_a = [1 Omega_c];

% 1. Impulse Invariance Method
[b_imp, a_imp] = impinvar(b_a, a_a, Fs);

% 2. Bilinear Transformation with Prewarping
Omega_p = (2/T)*tan(Omega_c*T/2);

b_pre = Omega_p;
a_pre = [1 Omega_p];

[b_blt, a_blt] = bilinear(b_pre, a_pre, Fs);

% Display coefficients
fprintf('IMPULSE INVARIANCE vs BILINEAR TRANSFORMATION\n\n');

fprintf('Cutoff Frequency = %.2f rad/s\n', Omega_c);
fprintf('Prewarped Frequency = %.2f rad/s\n\n', Omega_p);

disp('Impulse Invariance Numerator:');
disp(b_imp);

disp('Impulse Invariance Denominator:');
disp(a_imp);

disp('Bilinear Transformation Numerator:');
disp(b_blt);

disp('Bilinear Transformation Denominator:');
disp(a_blt);

% Frequency response
[H_imp, f] = freqz(b_imp, a_imp, 1024, Fs);
[H_blt, ~] = freqz(b_blt, a_blt, 1024, Fs);

% Plot magnitude and phase
figure;
```

```
subplot(2,1,1);
plot(f, abs(H_imp), 'b', f, abs(H_blt), 'r--');
grid on;
legend('Impulse Invariance', 'Bilinear Transformation');
title('Magnitude Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Magnitude');

subplot(2,1,2);
plot(f, angle(H_imp)*180/pi, 'b', ...
    f, angle(H_blt)*180/pi, 'r--');
grid on;
legend('Impulse Invariance', 'Bilinear Transformation');
title('Phase Response Comparison');
xlabel('Frequency (Hz)');
ylabel('Phase (Degrees)');
```

```
IMPULSE INVARIANCE vs BILINEAR TRANSFORMATION

Cutoff Frequency = 6283.19 rad/s
Prewarped Frequency = 7265.43 rad/s

Impulse Invariance Numerator:
    1.2566

Impulse Invariance Denominator:
    1.0000   -0.2846

Bilinear Transformation Numerator:
    0.4208    0.4208

Bilinear Transformation Denominator:
    1.0000   -0.1584
```



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